

Sample of Stack Exchange SQL Project

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How many users joined between 2020-2023?

```
# finding total sum of users joined since 2020
total_users_since_2020 <- dbGetQuery(db, 'SELECT SUM(Num_Users) AS Total_Users FROM (SELECT Count(*) AS
      FROM Users
      WHERE CreationYear >= "2020"
      GROUP BY CreationYear)')
total_users_since_2020
```

```
##      Total_Users
## 1          100796
```

100,796 users joined since 2020.

Verification

One way to verify that my result of total users joined since 2020 is correct is using the following query to display the the number of users who joined SO every year from 2020 to the latest year, which is 2023. The Count() function is used to count the total number of user IDs in the table, and using the WHERE clause to specify that we want the year of the CreationDate of the User account to be greater than 2020. The STRFTIME() function was also used to change the (%Y-%m-%d %H:%M:%S) format with the entire date and time to just (%Y) the year of the date. I then use GROUP BY to get the count of users for each year.

```
# selecting all columns with the creation date after 2020 (converting CreationDate to year from date-ti
# outputs number of users joined each year since 2020
users_joined_each_year <- dbGetQuery(db, 'SELECT Count(Users.Id) AS Num_Users, STRFTIME("%Y", CreationD
      FROM Users
      WHERE CreationYear >= "2020"
      GROUP BY CreationYear')
users_joined_each_year
```

```
# Using R to find the sum of the users joined each year found using the SQL query above
sum(users_joined_each_year$Num_Users)
```

```
## [1] 100796
```

How many answers are typically received before the accepted answer?

I interpreted this question as finding the dates that are received for an answer, before the accepted answer id and the sum of those answers for each question will be the number of answer posts that are received before the accepted answer. Here I am getting the dates for all questions that have an accepted answer, so the condition is that only question posts are extracted if the accepted answer id is not an empty value. As a result, this table contains the dates for all answers to a question that has an accepted answer id. It is used to find the number of answers typically received before the accepted answer.

##	Id	Question_PostDate	Answer_PostDate
## 1	1	2010-07-19 19:12:12	2010-07-19 19:19:46
## 2	1	2010-07-19 19:12:12	2010-07-19 20:23:57
## 3	1	2010-07-19 19:12:12	2010-07-19 22:40:47
## 4	1	2010-07-19 19:12:12	2010-07-20 05:13:21
## 5	1	2010-07-19 19:12:12	2010-09-15 21:08:26

Since I am already extracting the question posts and answer information for questions that have an accepted answer, then the number of answers received before (or other than) the accepted answer is the [total number of answers for a question - 1], which I am using the Count() function to calculate in this query, and also add another column displaying the values for each question post id.

##	Id	Question_PostDate	Answer_PostDate	NumAnswers_BeforeAccepted
## 1	1	2010-07-19 19:12:12	2010-07-19 19:19:46	5
## 2	2	2010-07-19 19:12:57	2010-07-19 19:24:35	6
## 3	3	2010-07-19 19:13:28	2010-07-19 19:14:43	18
## 4	4	2010-07-19 19:13:31	2010-07-19 21:31:53	4
## 5	7	2010-07-19 19:15:59	2010-07-19 19:18:41	24

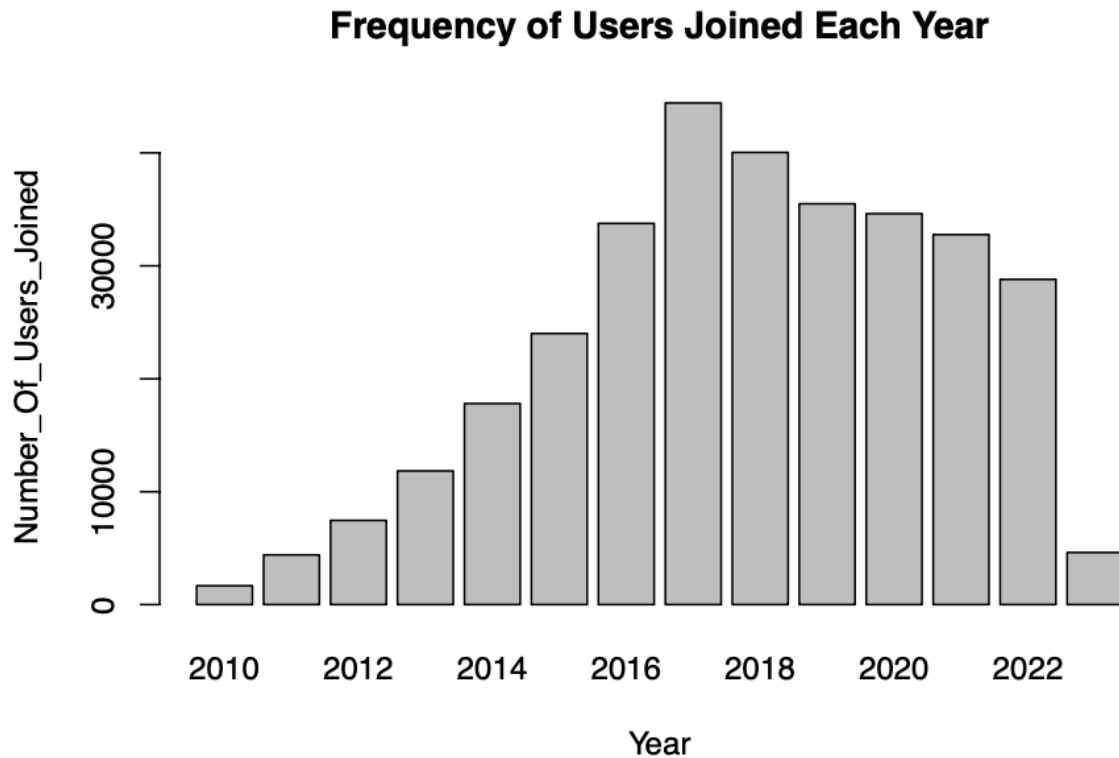
To find the typical value I am finding the average of the number of answers before the accepted answer for all the questions that have accepted answers.

```
dbGetQuery(db, 'SELECT AVG(NumAnswers_BeforeAccepted) AS AVG_NumAnsBeforeAA FROM
(SELECT Id, Question_PostDate, Answer_PostDate, (Count(Answer_PostDate)-1) AS NumAnswers_BeforeAccepted
(SELECT Q.Id, DATETIME(Q.CreationDate) AS Question_PostDate, DATETIME(A.CreationDate) AS Answer_PostDate
FROM Posts Q
INNER JOIN Posts A ON Q.Id=A.ParentId
WHERE Q.PostTypeId==1 and A.PostTypeId==2 and Q.AcceptedAnswerId != ""
ORDER BY Q.Id ASC, Answer_PostDate ASC)
GROUP BY Id)
')
```

```
## AVG_NumAnsBeforeAA
## 1 0.4762591
```

Finding the average of the column created in the table that displays the number of answers for each question before the accepted answer was 0.476. This means the average, or typical, number of answers received for a question post before an accepted answer is determined is approximately 0. This makes it evident that the accepted answer is often the first answer posted to a question.

How many users joined each year?



This bar plot displays the number of users who have joined Stack Overflow from 2010 to 2023, based on the results acquired from the SQL query above. There was a significantly low number of users who joined in 2010, although that could be explained due to Stack Overflow being a new resource and not as widely known. As it's popularity grew, it appears that the number of users who joined SO grew gradually from 2010, peaking at 2017 with a total of 44416 users, before starting to decline. From 2017 to 2022 the number of users who joined SO declined by a few thousand every year. It should be noted that the number of users joined in 2023 appears significantly low, however this number is only representative of the current time of this analysis, which is May, 2023.